Bump'em tower: printed parts

Profile: 90x90 light, B-type (Bosch-compatible), slot 10, 2 slots per face 45 mm apart

Fasteners: M6 ISO 7380 button head (4 mm hex) + existing slot-10 T-nuts; plates 5 mm thick

Material: PETG or ASA (not PLA: softens near warm drivers / supplies). Print flat, key side down; supports from

Device sizes: Futek IAA100 drawing FI1573, Mean Well RS-15 case 971A (docs/reference/).

48 V supply RSP-2000-48 (295x127x41 mm, 1.95 kg, case 952D): 2 printed end brackets, 10 mm air gap, ASA

Source: BumpemServer/hardware/enclosures (CadQuery). Dimensions in tower.py.

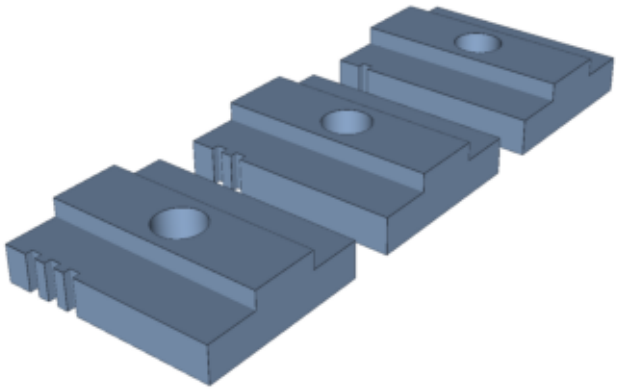
Part	Status	Still needed
Slot test plates	**READY TO PRINT**	nothing
Base mount plate	**READY (geometry)**	test-fit result (KEY_CLEAR), bolt length
IAA100 DIN-rail mount	**READY (datasheet)**	check your IAA100 has the DIN clip (66.8 x 50.4 x 17.8 mm version)
24 V supply plate (RS-15-24)	**READY (datasheet)**	M3 x 6 screws; a cover for the open 230 V terminals (electrician)
48 V supply brackets (RSP-2000-48)	**READY (datasheet)**	M4 x 12 screws (hole pattern checked against RSP-2000-SPEC p. 6)
Carrier board plate	**READY (KiCad)**	M3 x 16 + M3 nuts (4)
Ground bar plate (ONPIRA V 1x12)	**READY (listing)**	2 x M3 x 12 + 2 M3 nuts per plate
Signal connector plate	**CONCEPT**	MSTB panel part + cut-out size
Cable clip	**CONCEPT**	cable diameters (PLACEHOLDER 8 mm)
Power-strip hook (x2)	**CONCEPT**	power strip L x W x H

Order: 1) print the slot test plates, 2) report which fits, 3) measure the devices marked PLACEHOLDER, 4) final holders are generated from those numbers.

Mains: the power strip is only carried by printed hooks; no 230 V parts inside printed housings.

Slot test plates

READY TO PRINT



Role

Checks the fit before the real parts are printed: the key on the back must slide into the 10 mm profile slot without wobbling, the M6 bolt must pass the hole and still grip the T-nut.

How to anchor it

1. Slide a slot-10 T-nut into a free slot of the tower (or use one already in the slot).
2. Put the plate on the profile with the key (ridge on the back) in the slot.
3. Push an M6 button-head bolt through the hole and turn it into the T-nut, hand-tight.
4. Check: no sideways play, plate flat on the profile, at least ~5 mm of thread in the T-nut.
5. Report the result: too tight / good / loose (the 0.1 and 0.3 mm plates are in test_fit.stl).

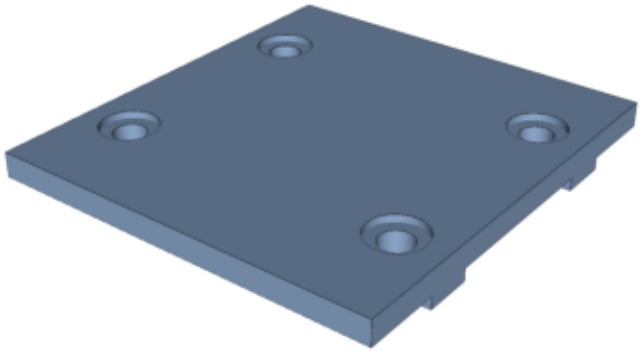
Data

Size 30 x 84 x 8 mm. Mount: M6 button-head bolts in slot-10 T-nuts, key 9.6 mm in the slot. Still needed: nothing.

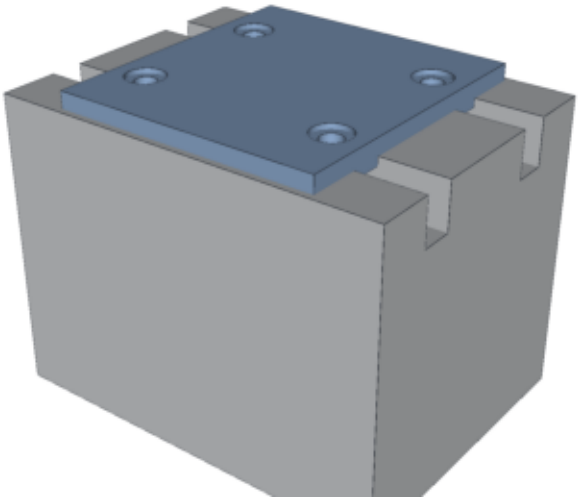
Base mount plate

READY (geometry)

Part



On the profile (grey), device in red



Role

Covers one face, 2 keys in both slots, 4 M6 holes with head seats. Template for wider parts.

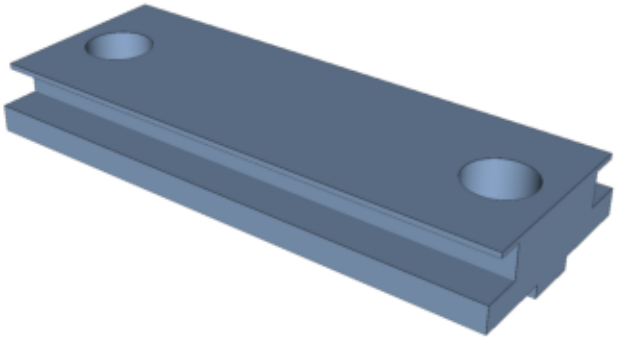
Data

Size 80 x 86 x 8 mm. Mount: M6 button-head bolts in slot-10 T-nuts, key 9.6 mm in the slot. Still needed: test-fit result (KEY_CLEAR), bolt length.

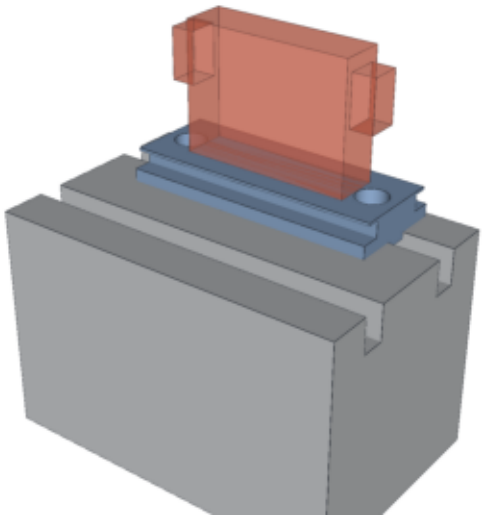
IAA100 DIN-rail mount

READY (datasheet)

Part



On the profile (grey), device in red



Role

Holds the Futek IAA100 force amplifier (now taped) on the tower. The IAA100 has a spring DIN clip on its back; this part is a short 35 mm DIN rail that bolts into one slot, so the amplifier clicks on and off without screws.

How to anchor it

1. Choose a slot on the tower face near the load cell cable; slide 2 slot-10 T-nuts into it, about 80 mm apart.
2. Put the mount on with the key in the slot and the rail facing out; bolt it with 2 M6 bolts through the holes at the ends (reach them through the openings in the rail).
3. Hook the top of the IAA100's DIN clip over the rail, press the bottom until it clicks.
4. Orientation: sensor plug (4-pin) toward the load cell, power/output plug (5-pin) toward the breadboard cable.
5. To remove: pull the clip's spring tab down (small screwdriver) and tilt the amplifier off.

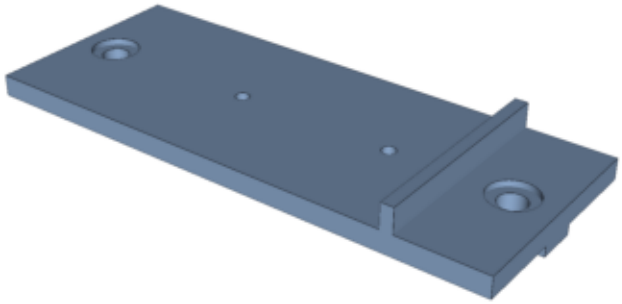
Data

Size 100 x 40 x 15 mm. Mount: M6 button-head bolts in slot-10 T-nuts, key 9.6 mm in the slot. Still needed: check your IAA100 has the DIN clip (66.8 x 50.4 x 17.8 mm version).

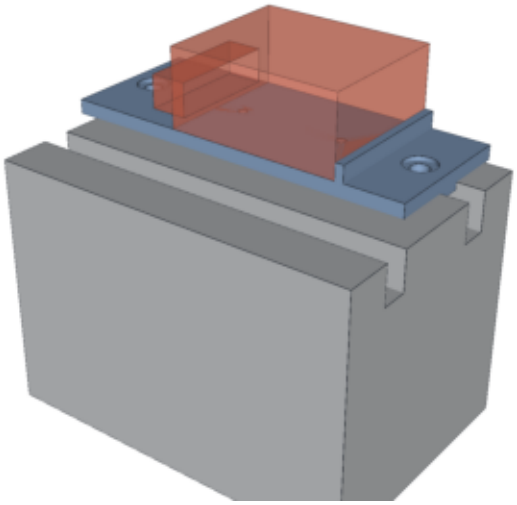
24 V supply plate (RS-15-24)

READY (datasheet)

Part



On the profile (grey), device in red



Role

Holds the Mean Well RS-15-24, the 24 V supply of the force amplifier (now taped). Fixed by the supply's own 2 M3 holes; an end stop keeps it square.

How to anchor it

1. Mains first: the supply stays unplugged; its 230 V terminals are wired and covered only by the electrician.
2. Put the supply on the plate, the end stop against the end opposite the screw terminals.
3. From the back of the plate, screw 2 M3 x 6 screws into the supply's bottom holes (max 3 mm into the case: no longer screws).
4. Slide 2 slot-10 T-nuts into a slot, 128 mm free; bolt the plate on with 2 M6 bolts through the holes at both ends (the one at the terminal end is 25 mm past the supply, away from the mains cables).
5. Route the 24 V output (+V, -V) to the IAA100 power plug (pin 1 +Vin, pin 2 GND).

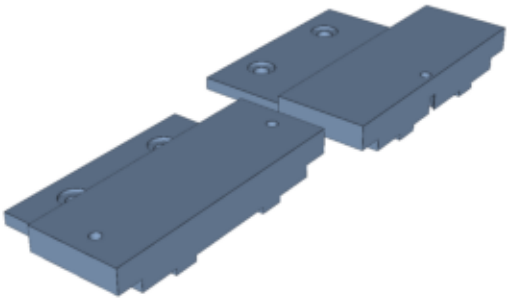
Data

Size 128 x 51 x 14 mm. Mount: M6 button-head bolts in slot-10 T-nuts, key 9.6 mm in the slot. Still needed: M3 x 6 screws; a cover for the open 230 V terminals (electrician).

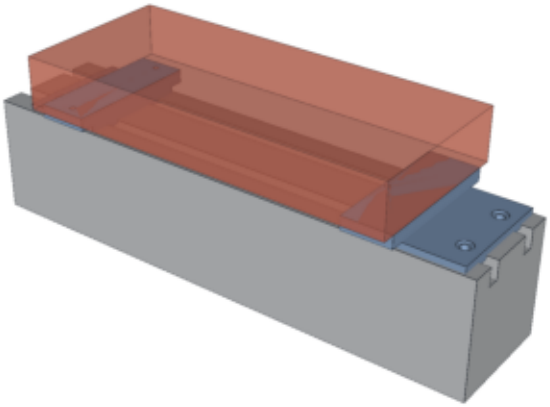
48 V supply brackets (RSP-2000-48)

READY (datasheet)

Part



On the profile (grey), device in red



Role

Hold the Mean Well RSP-2000-48, the motor supply (1.95 kg), on one profile face with a 10 mm air gap. Bracket A takes the 2 M4 holes at the end without fans, bracket B the single M4 hole at the fan end.

How to anchor it

1. Supply switched off and unplugged; leave its mains side to the electrician.
2. Lay the supply with its wide face (the one with 3 M4 holes, case 952D) up. Put bracket A on the end with 2 holes and bracket B on the fan end, pads on the supply, the bolt flanges pointing away from the supply.
3. Screw M4 x 12 screws from the back of each bracket into the supply (2 in A, 1 in B); max 4 mm into the case. B's screw goes into the hole near the middle of the fan end (34.5 mm from the end); the 2 small holes beside it are not for mounting.
4. Slide 4 slot-10 T-nuts into the tower face: one in each of its 2 slots for each bracket. The two brackets' bolts end up 339 mm apart.
5. Lift supply + brackets onto the profile, keys in both slots, and bolt each bracket with its 2 M6 bolts. Two people: one holds. Tighten firmly with the short end of the hex key only (about 2 N m): more crushes the plastic under the head.
6. Check: fans and vents free, 10 mm gap behind the supply, output cables (+V, -V) reach the chopper terminals 4 / 5.

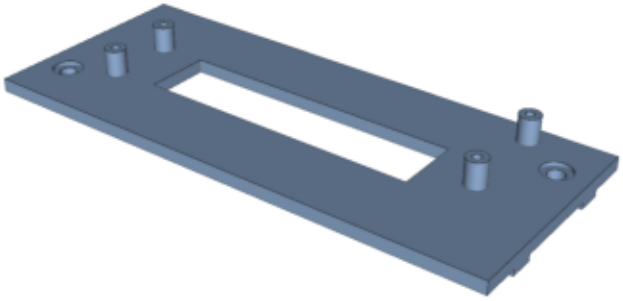
Data

Size 62 x 107 x 18 mm. Mount: M6 button-head bolts in slot-10 T-nuts, key 9.6 mm in the slot. Still needed: M4 x 12 screws (hole pattern checked against RSP-2000-SPEC p. 6).

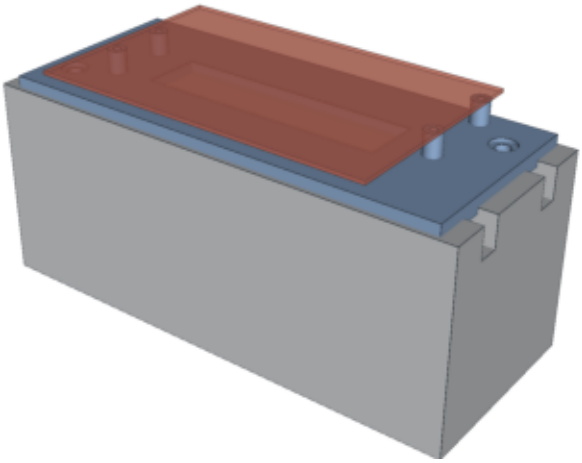
Carrier board plate

READY (KiCad)

Part



On the profile (grey), device in red



Role

Holds the carrier board (Teensy 4.1, DAC and the 4 module plugs) on one face of the tower that carries it, long side along the profile, on 10 mm standoffs so the soldered pins underneath touch nothing.

How to anchor it

1. Push an M3 nut into each of the 4 hexagon pockets in the back of the plate (a drop of glue or tape keeps them in).
2. Slide 2 slot-10 T-nuts into the face, one in each slot, 180 mm apart (the plate's 2 M6 holes sit diagonally).
3. Bolt the plate on, keys in both slots, with 2 M6 bolts; tighten with the short end of the hex key.
4. Put the board on the 4 standoffs, component side out: they only fit one way round (2 holes at one end, 2 at the other). Screw 4 M3 x 16 through the board into the nuts, gently: plastic underneath.
5. Teensy USB end up or down as the cables run best: turn the plate 180 degrees before bolting it on. Module plugs face one side: route the 4 module cables and the USB cable without strain.

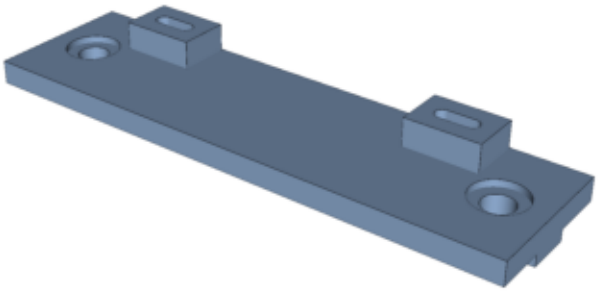
Data

Size 204 x 86 x 18 mm. Mount: M6 button-head bolts in slot-10 T-nuts, key 9.6 mm in the slot. Still needed: M3 x 16 + M3 nuts (4).

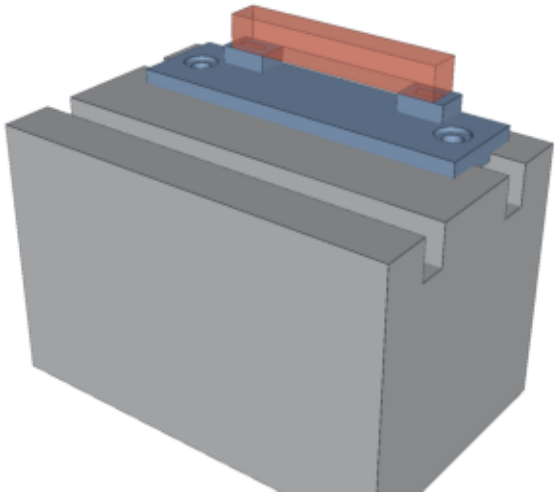
Ground bar plate (ONPIRA V 1x12)

READY (listing)

Part



On the profile (grey), device in red



Role

Holds the ONPIRA V 1x12 distributor: 12 screw terminals, all connected, so every ground wire of the tower (ESCON, IAA100, supplies' -V, module cable) lands on one bar instead of a twisted bundle.

How to anchor it

1. Slide an M3 nut into each of the 2 channels in the back of the plate (flat side up): it slides along, it cannot turn.
2. Put the board on the 2 pads (terminal screws up, beside the key, not over it). Line each nut up under the board's hole; push an M3 x 12 through the board into it and tighten gently. The included 5 mm sleeves and wood screws are not needed.
3. Slide 2 slot-10 T-nuts into a slot, 97 mm apart; bolt the plate on with 2 M6 bolts at its ends.
4. Wire the grounds; mark the bar GND. No 230 V protective earth on this bar (that stays with the electrician).

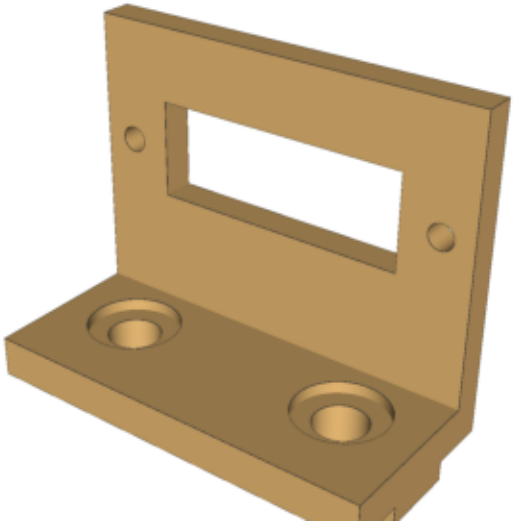
Data

Size 117 x 32 x 14 mm. Mount: M6 button-head bolts in slot-10 T-nuts, key 9.6 mm in the slot. Still needed: 2 x M3 x 12 + 2 M3 nuts per plate.

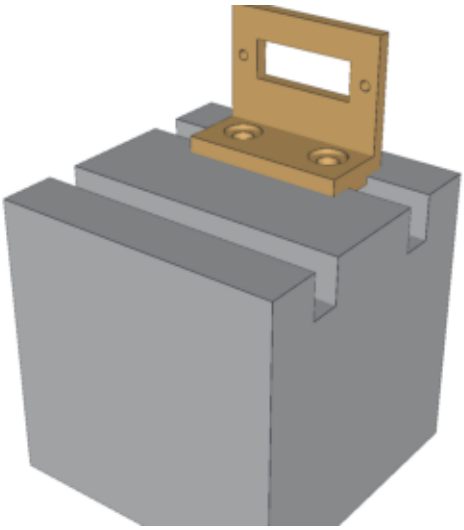
Signal connector plate

CONCEPT

Part



On the profile (grey), device in red



Role

Upright panel holding the tower's 5-pin MSTB socket: the single signal cable to the breadboard / carrier board.

Data

Size 50 x 24 x 43 mm. Mount: M6 button-head bolts in slot-10 T-nuts, key 9.6 mm in the slot. Still needed: MSTB panel part + cut-out size.

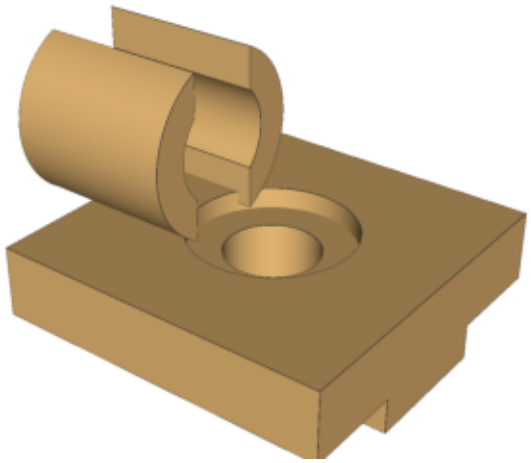
Note

Concept only: device sizes are placeholders, the final part will change.

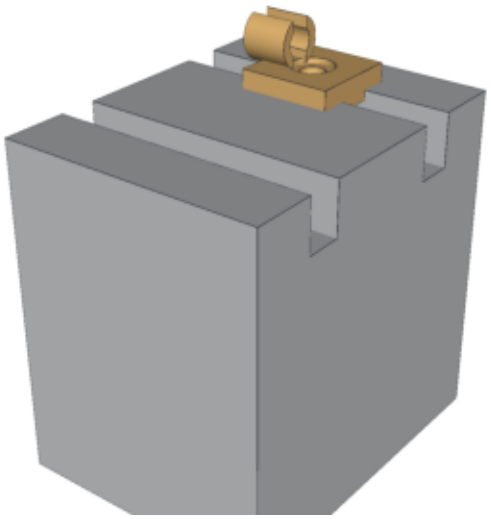
Cable clip

CONCEPT

Part



On the profile (grey), device in red



Role

Snap-in clip guiding a cable along the profile (mains, signal, motor).

Data

Size 28 x 24 x 18 mm. Mount: M6 button-head bolts in slot-10 T-nuts, key 9.6 mm in the slot. Still needed: cable diameters (PLACEHOLDER 8 mm).

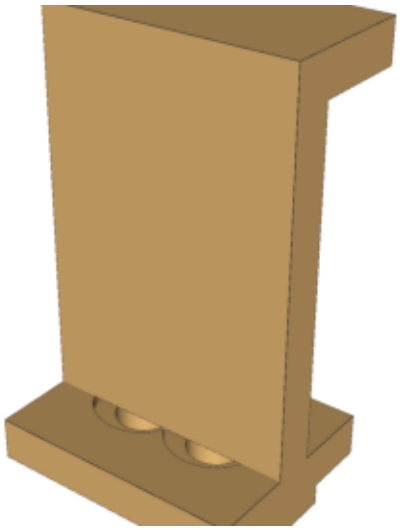
Note

Concept only: device sizes are placeholders, the final part will change.

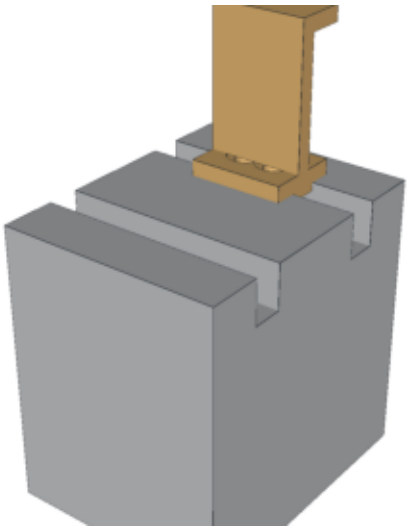
Power-strip hook (x2)

CONCEPT

Part



On the profile (grey), device in red



Role

Two hooks carrying the tower's power strip. Holder only, no mains parts inside the print.

Data

Size 30 x 26 x 53 mm. Mount: M6 button-head bolts in slot-10 T-nuts, key 9.6 mm in the slot. Still needed: power strip L x W x H.

Note

Concept only: device sizes are placeholders, the final part will change.